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COMPUTER

CHAPTER 11

Computer Network

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CHAPTER 11

Computer Network

Computer Awareness

IN THIS CHAPTER

- Network basics
- Network devices
- LAN, MAN and WAN
- Network topology
- Connection structures
- Revision facts

2.Assembly Language uses mnemonics.

3.Assembler converts Assembly Language into Machine Language.

4.FORTRAN was developed by John Backus in 1956.

5.C Language was developed by Dennis Ritchie in 1972.

6.C++ is an Object-Oriented Programming Language.

7.BCD uses 4 bits for each decimal digit.

8.ASCII uses 7 bits and represents 128 characters.

9.EBCDIC uses 8 bits and was developed by IBM.

10.1 Byte = 8 Bits.

A Computer Network is a group of two or more computers and other devices connected through communication channels for sharing data, resources, and information.

A network enables computers to:

- Communicate with each other
- Share files and data
- Share hardware resources (printers, scanners, etc.)
- Share software resources
- Access common services

Example

A school computer lab where all computers are connected to a central server is an example of a computer network.

BASIC COMPONENTS OF A COMPUTER NETWORK

1. Node (Station)

Every device connected to a network is called a Node or Station.

Examples

- Computer

- Server
- Router
- Switch
- Printer

Exam Point

Any device connected to a network = Node

2. Protocol

A Protocol is a set of rules used for communication between devices on a network.

Protocols ensure that data is transmitted and received correctly.

Examples

- TCP/IP
- HTTP
- FTP
- SMTP

Just as traffic rules control vehicles on roads, protocols control data transfer on networks.

3. Transmission Media

The physical path through which data travels is called Transmission Media. Examples

- Twisted Pair Cable
- Coaxial Cable
- Optical Fiber
- Wireless Signals

SERVER

A Server is a powerful computer that provides resources and services to other computers connected to a network.

It stores:

- Data
- Programs
- Files
- Applications

Characteristics

- High processing speed
- Large storage capacity
- Large RAM
- Serves multiple users simultaneously

Example

A college server stores student records that can be accessed by authorized computers.

Exam Point

Server provides resources; Client uses resources.

TERMINAL

A Terminal is a device consisting of a keyboard and display screen used to communicate with a computer.

Types of Terminals

1. Dumb Terminal

2. Smart Terminal

3. Intelligent Terminal

4. 3270 Terminal

- IBM display terminal
- Used to communicate with IBM mainframe computer

NETWORK DEVICES

Network devices are hardware components that help computers communicate in a network.

1. REPEATER A Repeater is a device that regenerates weak signals into strong signals and extends the transmission distance.

Characteristics

- Works at Physical Layer
- Regenerates signals
- 2-port device

Function

Weak Signal → Repeater → Strong Signal

2. HUB

A Hub is a multi-port repeater.

Characteristics

- Connects multiple devices
- Sends data to all connected devices
- No filtering capability
- Works at Physical Layer

Exam Point

Hub broadcasts data to all devices.

3. BRIDGE

A Bridge connects two LANs and filters data using MAC addresses.

Characteristics

- Works at Data Link Layer

- Filters traffic
- Uses MAC Address
- 2-port device

Exam Point

Bridge = Connects two LANs

4. SWITCH

A Switch is an advanced multi-port bridge.

Characteristics

- Works at Data Link Layer
- Sends data only to intended device
- Performs error checking
- Improves network performance

Advantages

- Less traffic
- Faster communication
- More efficient than Hub

Exam Point

Switch forwards data only to the correct destination port.

5. ROUTER

A Router forwards data packets using IP addresses.

Characteristics

- Works at Network Layer
- Connects LANs and WANs
- Uses Routing Table
- Selects best path for data

Example

Home Wi-Fi Router

6. GATEWAY

A Gateway connects two different networks which are using different protocols.

Exam Point

Gateway = Protocol Converter

7. BROUTER

Brouter (Bridge + Router) combines features of both bridge and router.

Functions

- Routes packets like a router
- Filters traffic like a bridge

8. NIC (Network Interface Card)

NIC is a hardware device that connects a computer to a network.

Characteristics

- Unique MAC Address
- Installed inside computer
- Connects LAN

Exam Point NIC contains MAC Address.

9. MODEM (MOdulator DEModulator)

A Modem connects a computer network to the Internet.

Modulation :Converts Digital Signal into → Analog Signal

Demodulation :Converts Analog Signal → Digital Signal

Exam Point

Modem = Signal Translator

TYPES OF COMPUTER NETWORKS

Computer Networks

- LAN
- MAN
- WAN

1. LAN (LOCAL AREA NETWORK)

A LAN is a network that covers a small geographical area.

Examples

- Home Network
- School Network
- Office Network

Characteristics

- High speed
- Low cost
- Covers few kilometers
- Data Rate: 100 Mbps to 1000 Mbps

Exam Point

LAN = Small Area

2. MAN (METROPOLITAN AREA NETWORK)

MAN covers a city or metropolitan area.

Characteristics

- Larger than LAN
- Smaller than WAN
- Covers 5–50 km

Example

City-wide Airtel Network

3. WAN (WIDE AREA NETWORK)

WAN covers a very large geographical area.

Examples

- Internet
- Banking Networks
- Multinational Companies

Characteristics

- Covers countries and continents
- Uses satellite, fiber optic and telephone lines

Exam Point

Internet is the largest WAN.

Memory Trick

LAN → Building

MAN → City

WAN → World

NETWORK PERFORMANCE TERMS

Transmit Time

The time required to send data from one device to another.

Response Time

The time between sending a request and receiving a response.

Example

Clicking a website link and waiting for the webpage to open.

NETWORK TOPOLOGY

Topology refers to the physical arrangement of computers and devices in a network.

TYPES OF TOPOLOGY

Network Topology

- Bus
- Ring
- Star
- Mesh
- Tree

1. BUS TOPOLOGY



Bus: computers share a backbone connection.

Structure: (Single Backbone Cable)

Characteristics

- Single central cable
- Data travels in both directions
- Easy installation
- Low cost

Advantage

- Easy to set up.
- Disadvantage
- Failure of backbone cable affects entire network.

Exam Point Bus Topology uses a Backbone Cable.

2. RING TOPOLOGY

Structure

Characteristics

- Circular arrangement
- Data travels in one direction
- No central device

Advantage

- Equal access for all nodes.

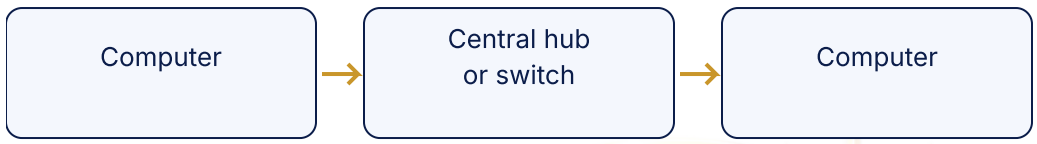
Disadvantage

- Failure of one node can affect communication.

Exam Point

Ring = Closed Loop Network

3. STAR TOPOLOGY



Star: devices connect through a central node.

Structure:

Characteristics

- All nodes connect to central device
- Easy to install
- Easy troubleshooting

Advantage

- Failure of one cable affects only one computer.

Disadvantage

- Failure of central hub/switch affects entire network.

Exam Point

Most commonly used topology = Star Topology

4. MESH TOPOLOGY

Structure

Every device is connected to every other device.

Characteristics

- Dedicated point-to-point connections
- Highly reliable
- Very secure

Uses

Military communication systems

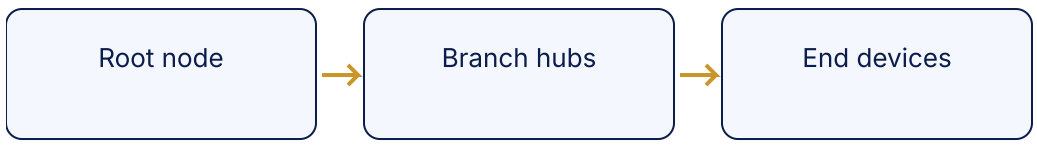
Advantage

Multiple connections provide redundancy; failures can still interrupt communication if enough links or nodes fail.

Exam Point

Full mesh provides many alternative paths and high redundancy; reliability still depends on the design and equipment.

5. TREE TOPOLOGY



Tree: a hierarchy of connected nodes.

Tree Topology is a combination of Star and Bus Topology.

Characteristics

- Hierarchical structure
- Scalable
- Hybrid topology

Exam Point

Tree Topology = Star + Bus

KEY DIFFERENCES

Hub vs Switch

Feature	Hub	Switch
Intelligence	No	Yes
Data Forwarding	All Devices	Specific Device
Speed	Slow	Fast
Error Checking	No	Yes

Bridge vs Router

Feature	Bridge	Router
Address Used	MAC Address	IP Address
OSI Layer	Data Link	Network
Connects	LANs	LANs and WANs

LAN vs MAN vs WAN

Feature	LAN	MAN	WAN
Area Covered	Small	City	Large
Range	Few km	5–50 km	50 km+
Speed	Highest	High	Lower
Example	School	City Network	Internet

MOST IMPORTANT FACTS FOR EXAMS

1.Protocol = Set of communication rules.

2.Server provides resources to clients.

3.Repeater regenerates signals.

4.Hub is a multi-port repeater.

5.Bridge uses MAC addresses.

6.Switch is a multi-port bridge.

7.Router uses IP addresses.

8.Gateway is a protocol converter.

9.NIC contains MAC address.

10.Modem = Modulator + Demodulator. 11.LAN covers small areas.

12.MAN covers a city.

13.Internet is the largest WAN.

14.Star topology is the most commonly used topology.

15.Mesh topology is the most reliable topology.

QUICK REVISION NOTES

- Protocol = Rules for communication
- Server = Provides resources
- Repeater = Regenerates signal
- Hub = Multi-port repeater
- Bridge = Uses MAC address
- Switch = Smart bridge
- Router = Uses IP address
- Gateway = Protocol converter
- NIC = Network adapter
- Modem = Converts Analog ↔ Digital
- LAN = Small area

- MAN = City area
- WAN = Large area
- Internet = Largest WAN
- Star = Most popular topology
- Mesh = Most reliable topology
- Tree = Star + Bus

COMPUTER NETWORK ARCHITECTURES

A Network Architecture refers to the overall design and structure of a computer network. It defines how computers communicate, share resources, and exchange information with each other.

TYPES OF NETWORK ARCHITECTURES

Computer Network Architectures

- Client-Server Architecture
- Peer-to-Peer (P2P) Architecture
- Hybrid Architecture

1. CLIENT-SERVER ARCHITECTURE

In Client-Server Architecture, one or more powerful computers act as Servers, while other computers act as Clients.

The server provides resources and services, and clients request those resources.

Components

1.Server

A central and powerful computer that: